

CHICAGO EXECUTIVE, IL, USA

WMO: 744665

Lat: 42.121N

Lon: 87.905W

Elev: 194

StdP: 99.02

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04838

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.3	-15.8	-24.1	0.4	-17.5	-21.8	0.5	-14.8	10.9	-1.1	9.6	-1.1	3.9	270	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	33.0	23.3	31.6	22.6	30.1	21.9	24.9	30.7	24.0	29.3	23.1	28.1	5.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.1	28.2	22.3	17.3	27.3	21.4	16.4	26.4	76.3	30.9	72.7	29.3	69.5	28.3	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.6	8.5	7.7	-21.9	35.5	3.8	1.8	-24.7	36.8	-26.9	37.9	-29.1	38.9	-31.8	40.3		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-22.3	26.2	3.5	0.8	-24.8	26.8	-26.9	27.3	-28.9	27.8	-31.5	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	-4.2	-3.0	3.3	9.3	14.8	20.7	23.6	22.7	18.9	11.9	5.1	-0.9
	DBStd	10.92	6.56	6.05	6.09	5.16	5.14	4.12	3.38	3.00	4.25	4.85	5.21	5.69
	HDD10.0	1649	442	366	224	73	11	0	0	0	35	160	339	
	HDD18.3	3446	700	598	467	275	136	22	2	43	208	396	596	
	CDD10.0	1744	1	1	17	52	161	322	421	393	268	93	14	1
	CDD18.3	498	0	0	2	4	28	93	164	138	61	8	0	0
	CDH23.3	4474	0	0	14	60	296	909	1548	1074	508	64	0	0
	CDH26.7	1522	0	0	2	12	88	326	578	342	162	13	0	0
Wind	WSAvg	3.5	4.1	4.0	3.9	4.1	3.5	3.1	2.9	2.6	2.8	3.4	3.9	3.9
Precipitation	PrecAvg	932	47	39	69	98	90	102	99	91	100	65	72	63
	PrecMax	1207	131	90	134	159	210	204	283	198	353	202	180	142
	PrecMin	679	7	5	9	30	36	17	33	21	0	6	13	11
	PrecStd	137	30	22	30	38	40	44	50	58	77	42	37	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	16.1	25.3	28.3	31.6	34.1	36.0	34.0	32.9	28.4	20.4	16.0
		MCWB	11.8	11.3	17.2	18.0	20.9	23.4	24.4	23.4	23.1	20.8	14.3	13.5
	2%	DB	9.9	11.9	19.9	24.9	29.2	32.4	33.3	32.0	30.8	25.0	17.6	12.1
		MCWB	7.0	8.1	14.0	16.1	19.8	22.6	23.8	23.0	21.9	18.2	13.3	9.8
	5%	DB	6.4	7.6	16.2	22.0	26.8	30.3	31.6	30.2	28.3	22.5	15.8	9.0
		MCWB	4.0	4.7	11.6	14.3	18.7	21.8	23.1	22.2	20.6	17.0	12.0	6.8
	10%	DB	4.1	5.0	12.6	18.9	24.1	28.2	29.9	28.4	26.4	19.9	13.2	7.1
		MCWB	2.5	2.7	8.5	12.5	17.8	20.8	22.3	21.4	19.6	15.4	9.8	5.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.7	12.6	17.9	19.2	22.9	25.4	26.3	25.2	24.2	21.9	16.6	13.8
		MCDB	13.7	15.0	24.6	26.0	29.4	32.0	33.1	31.6	30.4	27.2	18.5	15.6
	2%	WB	7.6	8.3	15.0	17.1	21.2	23.8	25.2	24.2	23.0	19.9	14.4	10.0
		MCDB	9.3	11.4	18.9	23.3	26.9	29.5	30.9	29.2	28.1	22.9	17.1	11.5
	5%	WB	4.5	5.1	12.2	15.3	19.9	22.8	24.3	23.4	21.9	17.9	12.1	7.2
		MCDB	6.1	7.7	15.9	20.9	25.1	28.7	29.9	28.0	26.9	21.4	14.6	8.6
	10%	WB	2.6	2.8	8.5	13.0	18.4	21.7	23.4	22.5	20.7	15.9	10.1	4.9
		MCDB	4.3	4.7	12.7	18.9	23.2	27.1	28.4	26.9	25.1	19.3	13.2	6.7
Mean Daily Temperature Range	5% DB	MDBR	7.5	8.0	8.9	10.2	10.6	10.4	9.6	9.3	10.0	9.2	8.2	6.9
		MCDBR	9.7	11.2	13.5	15.7	14.1	12.4	11.1	11.4	12.2	12.7	11.1	9.4
		MCWBR	8.2	8.5	9.2	9.2	7.3	5.8	4.7	4.8	5.5	7.3	8.0	7.7
	5% WB	MCDBR	9.2	10.9	12.7	14.0	12.4	11.3	10.0	9.7	10.6	10.5	9.8	8.6
		MCWBR	8.2	8.7	9.3	9.4	7.2	6.1	5.1	4.9	5.4	7.5	8.2	8.0
Clear-Sky Solar Irradiance	taub	0.297	0.312	0.344	0.399	0.426	0.439	0.443	0.437	0.408	0.355	0.314	0.296	
	taud	2.441	2.392	2.390	2.260	2.240	2.252	2.240	2.255	2.305	2.454	2.515	2.477	
	Ebn at Noon	853	894	901	869	852	839	830	823	821	832	827	822	
	Edh at Noon	76	94	106	130	137	135	136	130	115	88	71	68	
All-Sky Solar Radiation	RadAvg	1.55	2.43	3.57	4.61	5.28	6.05	6.11	5.31	4.37	2.85	1.80	1.29	
	RadStd	0.18	0.30	0.29	0.38	0.42	0.41	0.29	0.32	0.37	0.32	0.18	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page